



SMART ATTENDANCE SYSTEM USING FACIAL RECOGNITION

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Abstract

- Smart attendance system for proxy-resistant attendance.
- Uses face detection, liveness and session verification.
- Reduces proxy attendance and manual effort.

Problem Statement & Objectives

- Problem: Traditional attendance is vulnerable to proxy attendance.
- Objective 1: Verify student presence
- Objective 2: Reduce proxy attendance
- Objective 3: Prevent multiple-person attendance
- Scope: Faculty, Students | Attendance, Verification | Classroom-based

Existing vs Proposed System

Feature	Existing System	Proposed System
Attendance	Manual	Automated
Verification	Basic	Multi-factor
Proxy	Possible	Reduced
Records	Manual	Digital
Time Control	No	Time-Locked

Feasibility Study

- Technical: Uses web, Camera, MongoDB
- Economic: Low-cost , open-source technologies
- Operational: Simple faculty & Student workflow
- Schedule: Can be developed in modules within the project timeline

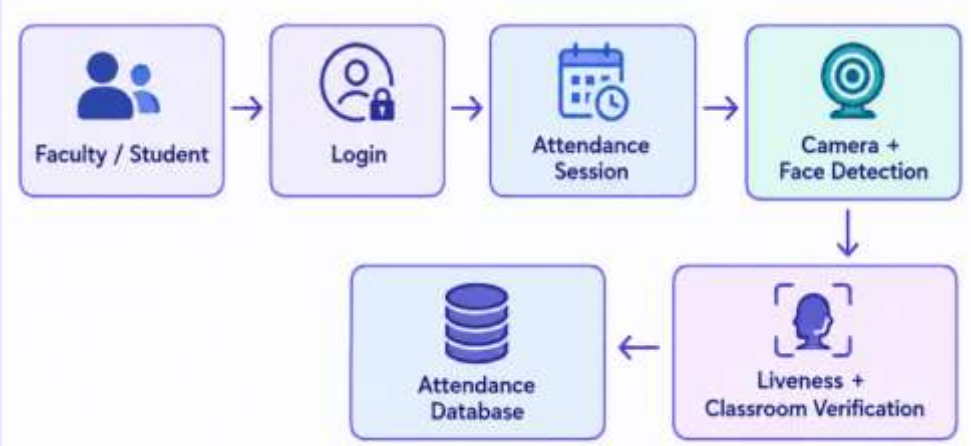
Results & Testing

Valid login → **Pass**
Single face → **Attendance allowed**
Multiple faces → **Blocked**
Failed liveness → **Blocked**
Valid session → **Attendance recorded**

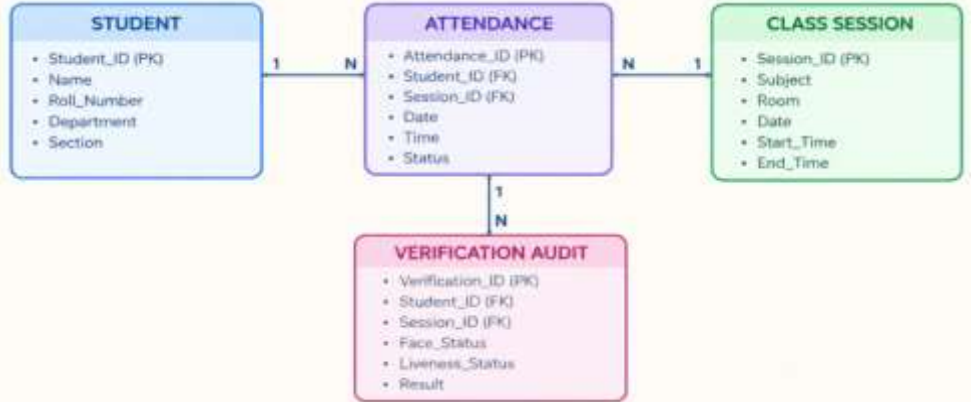
Tools & Technologies

- Front-End: HTML, CSS, JavaScript
- Back-End: Node.js / REST APIs
- Database: MongoDB
- Tools: Git & GitHub, Postman, VS Code
- Others: Web camera, Face Detection

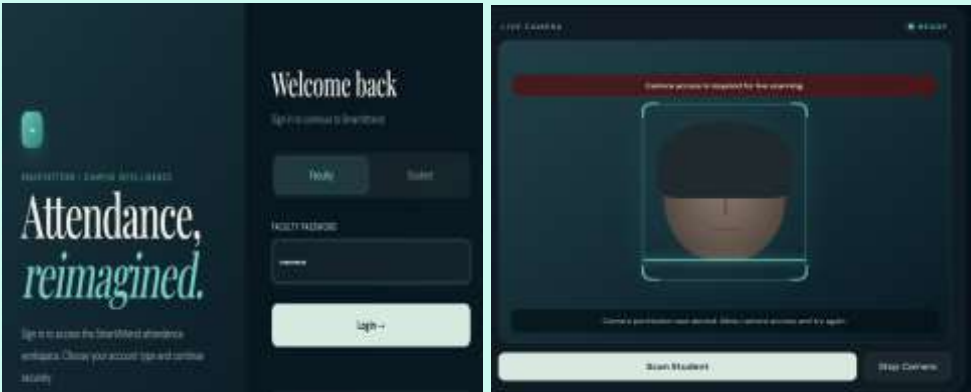
System Architecture



ER Diagram & Database Design



Implementation Screenshots



Conclusion

- Automates attendance
- Reduces proxy attendance
- Improves verification
- Maintains digital records

Future Scope

- Real-time face recognition
- Cloud MongoDB
- Mobile application
- ERP integration

